

Problem 5.12

Suppose that the interest earned using the Banker's Rule is \$40.58. What is the interest earned using the exact simple interest method?

Let I_b be the interest using Banker's Rule and I_e the interest using exact simple interest.

We know that

$$I_b = Pr \left(\frac{t}{360} \right)$$

and

$$I_e = Pr \left(\frac{t}{365} \right).$$

From the previous relationship,

$$\frac{I_e}{I_b} = \frac{360}{365}.$$

Since $I_b = 40.58$, we have

$$I_e = 40.58 \left(\frac{360}{365} \right).$$

Therefore,

$$I_e \approx 40.02.$$

So, the interest earned using the exact simple interest method is

$$\boxed{\$40.02}.$$